

ForMoSA: Forward Modeling tool for Spectral Analysis

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Summary

ForMoSA (FORward MOdeling tool for Spectral Analysis) is an open-source `Python` package designed to fit spectroscopic and photometric observations using a Bayesian framework. It has been mainly designed for fitting directly imaged young planetary-mass brown dwarfs and exoplanets, and it can utilize different self-consistent atmospheric models to perform robust parameter space exploration. The developments within **ForMoSA** are supported by an international collaboration among several laboratories in France (IPAG, LIRA, LAM, and Lagrange), Germany (MPIA), USA (NASA Goddard), and Chile (FCLA, Universidad Diego Portales, and Universidad de Chile). The evolution of the code and the growing interest from the scientific community has led to the need for this dedicated publication. This paper is published alongside the release of **ForMoSA v2.0**, which has been refactored into a class architecture with user-friendly features and new functionalities.

Statement of need

Recent advances in ground- and space-based observatories now enable routine, high-quality observations of exoplanet atmospheres across a wide range of wavelengths and resolutions. In practice, **ForMoSA** was designed to bridge the gap between these observations and atmospheric models by providing a Bayesian framework to robustly compare the two. **ForMoSA** adopts a forward-modeling approach based on exploring through a nested sampling (Skilling 2004) the parameter space of pre-computed, self-consistent atmospheric model grids.

The code **ForMoSA** was developed to address the growing need for an efficient,

standardized framework to analyze these high-quality observations, and addresses the following needs:

- **Computational efficiency:** It leverages `xarray`¹, enabling fast interpolation and manipulation of large model grids.
- **Data modularity:** It allows the simultaneous analysis of diverse datasets, even when they overlap in wavelength coverage. This includes multiple datasets taken at different times and with different setups.
- **Flexible likelihood mappings:** Users can choose from various likelihood metrics to account for specific observational uncertainties (Ruffio et al. 2019).
- **Extensive parameterization:** Beyond the atmospheric model parameters, a comprehensive suite of extra-grid parameters, such as radial velocity, rotational broadening, and scaling factors, can be incorporated into the fit.
- **High-Contrast module:** It incorporates the method of (Landman et al. 2024), allowing users to include reference stellar, atmospheric transmission, and systematics spectra to accurately fit contrast-limited data.

State of the field

The landscape of exoplanetary atmospheric modeling is broadly divided into two categories: “free retrieval” frameworks, which compute radiative transfer on the fly to fit parameterized atmospheric structures, and “forward modeling” codes, which interpolate pre-computed grids of self-consistent physical models.

The codes in the first category are **CHIMERA** (Line et al. 2013), **petitRADTRANS** (Mollière et al. 2019), **NEMESIS** (Irwin et al. 2008), **CROCODILE** (Hayoz et al. 2023), and **Pyrat Bay** (Cubillos and Blečić 2021). These tools are very flexible but also computationally expensive. Other specialized tools can handle specific niche problems, such as opacity management (**Exo_k**, Leconte 2021), high-resolution autodifferentiation (**ExoJAX**, Kawahara et al. 2022), phase curves (**APOLLO**, Howe, Burrows, and Deming 2017), 1D disequilibrium chemistry (**HOMER**, Himes et al. 2022), or rocky planet emission (**rfast**, Robinson 2023).

ForMoSA complement these on-the-fly radiative transfer or free-retrieval codes by offering a faster, grid-interpolated alternative for parameter estimation. **ForMoSA**’s closest existing counterparts are **species** (Stolker et al. 2020) and **SEDA** (Suarez et al. 2025), which also facilitate Bayesian inference between spectro-photometric observations and pre-computed atmospheric grids. **species** is a generalized, all-in-one toolkit for direct imaging that encompasses everything from flux calibration to color-magnitude diagrams. **SEDA** is more similar to **ForMoSA**, but it was only introduced early 2025 to the

¹`xarray` documentation: <https://docs.xarray.dev/en/stable>

community, and has not yet been widely adopted. We chose to build **ForMoSA** as a standalone package to provide a highly specialized, modular inversion framework expressly optimized for data heterogeneity and statistical flexibility. Integrating these new capabilities into existing tools would have required fundamental architectural changes.

Software design

The Python-based code of **ForMoSA** is organized into several modules, as shown in Figure 1. In practice, **ForMoSA** v2.0 has a class-based architecture, with some main modules (config, observation, filter, grid, parameter, nested_sampling), and support modules (core, utils, and transform). Because **ForMoSA** is an end-to-end Bayesian tool that handles all stages of the analysis, its installation requires several **Python** packages, as well as downloading the model grids needed for the fit. All information required for installation, access to available grids, and the tutorials material is available at the **ForMoSA** documentation².

In summary, to start an analysis, the user must first provide an atmospheric model grid and the observations. Then, the user must provide a `.ini` configuration file containing all of **ForMoSA**'s input parameters, including the paths, grid adaptation keys, nested sampling keys, and the free parameters priors configuration. **ForMoSA**'s main modules are centralized through the Analysis class, which creates and handles sub-grids of atmospheric models tailored to each observation, before launching the nested sampling algorithm using the support modules. At each iteration of the nested sampling, random parameter values are drawn from the prior distributions. The sampled parameters are used to interpolate model spectra from the sub-grids, before comparing them to each observation by computing the log-likelihood function. For a given run, the data, sub-grids, parameters, and results are automatically saved to paths specified by the user, and can be easily recovered for subsequent analysis and visualized through the Plotting class.

Research impact statement

Initiated in 2020, **ForMoSA** has become a highly adaptable tool for the atmospheric characterization of substellar objects. Its computational efficiency and compatibility with diverse physical models have enabled the standard analysis of individual targets across a wide range of spectral resolutions, from broadband photometry to high-resolution spectroscopy. Furthermore, its speed allows for the homogeneous analysis of large spectral libraries. By applying **ForMoSA** to extensive populations of brown dwarfs and planetary-mass companions, researchers have successfully identified structural systematics in atmospheric grid models, such as incomplete treatments of clouds and dust (e.g., Petrus et al. 2020, 2025; Palma-Bifani et al. 2025).

²**ForMoSA** documentation: <https://ForMoSA.readthedocs.io/en/latest/index.html>

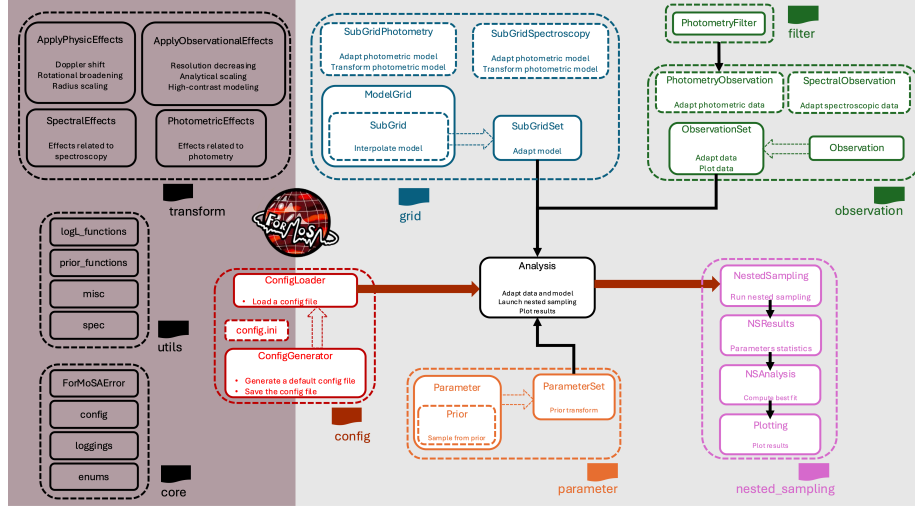


Figure 1: **ForMoSA** workflow diagram. The shaded gray area on the right represents the core functionalities required to run the Nested Sampling. The left dark-gray area represents utility and support functions. The small boxes contain the modules, and the dashed boxes represent the contents of each folder.

A major strength of **ForMoSA** lies in its capacity to jointly fit multi-instrument, heterogeneous datasets to maximize wavelength coverage. Because combining data from diverse instruments often introduces biases, **ForMoSA** features the dedicated MOSAIC module. This framework utilizes customized likelihood scaling and parametric covariance models to mitigate inter-calibration offsets and complex noise structures. This robust statistical approach has been critical for accurately characterizing complex benchmark companions like HIP 65426 (Petrus et al. 2021; Carter et al. 2023), AB Pic b (Palma-Bifani et al. 2023), VHS 1256 b Radcliffe et al. (2026), AF Lep b (Palma-Bifani et al. 2024), β Pic b (Houll   et al. 2025; Ravet et al. 2025), YSES 1 b and c (Hoch et al. 2025), and COCONUTS-2 b (Ravet et al. 2026), preventing the misinterpretation of atmospheric properties that can arise from dataset discrepancies. Finally, the modularity of **ForMoSA** include a high-contrast, high-resolution (HCHR) module applied to VLT/HiRISE data ($R > 100,000$) to extract precise radial velocities and detailed orbital constraints (Denis et al. 2025, 2026). To track the ongoing scientific output associated with the tool, a continually updated list of peer-reviewed publications utilizing **ForMoSA** is maintained on our NASA ADS public library³.

ForMoSA is an active project that has been used in multiple published projects of research. It will continue to evolve over time, with new functionalities be-

³NASA ADS Library: <https://ui.adsabs.harvard.edu/user/libraries/PekELjOGR4yl3XOnGwOAng>

ing added to meet the requirements of future instruments (ELTs, HWO, etc.), updates to the included model grids, and the coupling of atmospheric models with additional physics to better fit the data (e.g., disks, extinction laws, multiple columns, time variability). For this reason, the official documentation at ReadTheDocs, which will be regularly updated, should be considered the reference description of the code.

AI usage disclosure

Our team fully designed and authored the core architecture, logic, and implementation of the code without AI generation. We used AI strictly as an auxiliary tool for debugging and managing documentation. Additionally, we utilized Gemini to refine the English and Gemini coupled with Antigravity to format and compile the main text and the bibliography.

Acknowledgements

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Appendix A: Performance

The computational cost of **ForMoSA** is primarily driven by the number of forward model evaluations required for the nested sampling algorithm to converge. While absolute execution time (inversion time) depends on the specific hardware, Figure 2 provides a comparative analysis to guide users in defining the configuration file. The total inversion time is a multi-dimensional function depending on several factors, including the spectral resolution (R_λ), the signal-to-noise ratio (S/N), the number of live points, the dimensionality of the parameter space, and the machine used.

In the left panel of Figure 2, we observe that the main driver of computational cost is indeed the spectral resolution. As the spectral resolution increases, the

⁴Code/Astro Workshop: <https://semaphore.github.io/codeastro/>

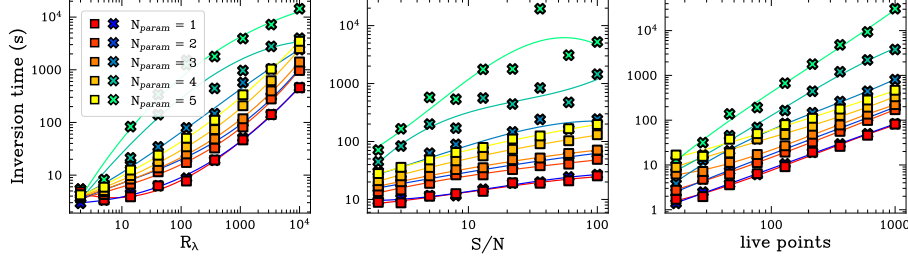


Figure 2: Performance comparison between the nested sampling algorithms `PyMultiNest` (squares) and `Nestle` (crosses). Different colors show the number of free parameters used (from 1 to 5). From left to right: Inversion time as a function of spectral resolution (R_λ), signal-to-noise (S/N), and number of live points. The default setup is $R_\lambda = 368$, $S/N = 22$, and 215 live points.

number of data points also increases, and the number of points to be compared in the likelihood function grows, directly increasing the computational cost of each forward model call. In the middle panel, we observe the dependence on the S/N. Here, noise is applied using the following formula:

$$\vec{n} \sim \mathcal{N}(0, \sigma^2), \quad \text{with} \quad \sigma = \frac{\text{mean}(d)}{S/N},$$

where d is the synthetic flux. As the S/N increases, the error bars become smaller, which leads to a more sharply peaked likelihood function. This tighter posterior distribution requires the sampler to explore the parameter space with finer precision and typically increases the number of likelihood evaluations, thereby lengthening the total inversion time. In the right panel, we observe that the inversion time scales approximately linearly with the number of live points. The different colors in Figure 2 indicate how the dimensionality of the parameter space (number of free parameters) strongly drives the inversion time. As the complexity of the model increases, the “volume” of the prior space grows, requiring more iterations to isolate the high-likelihood regions. In addition, we also provide a comparison between `PyMultiNest` (squares) and `nestle` (crosses). While `nestle` is a convenient pure-Python implementation, `PyMultiNest` (leveraging the Fortran `MultiNest` library) generally demonstrates superior scaling and efficiency.

To optimize the integration time of a **ForMoSA** inversion, we advise users to:

- Start low and run initial tests with a reduced number of live points (<100) to verify the model setup.
- Use a spectral resolution that matches the physical information content of the data; over-sampling the spectrum increases the inversion time

without improving accuracy.

- Select the right algorithm for the inversion. For high-dimensional fits (>5 parameters), **PyMultiNest** is the recommended default.

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